

Aditya Singh Baghel

Bhopal, MP, India | +91-9340846607 | adityasinghbaghel678@gmail.com | LinkedIn | GitHub | LeetCode

PROFESSIONAL SUMMARY

Aspiring Data Scientist with hands-on experience in statistical modeling, machine learning, deep learning, and data pipeline development. Built end-to-end solutions including an ECG stress classifier (CNN-BiLSTM, 92%+ accuracy) and a real-time YOLO-based traffic monitoring system. Proficient in Python, SQL, TensorFlow, Power BI, and Tableau. Skilled in feature engineering, NLP, Big Data concepts, and cross-functional collaboration. CGPA: 9.3/10. Certified in ML (DeepLearning.AI) and SQL.

EDUCATION

SAGE University Bhopal (Sunstone)

B.Tech – Computer Science Engineering, Specialization: Data Analytics | **CGPA: 9.3 / 10.0**

Model Hr. Sec. School Civil Lines, Rewa

Class XII – Science Stream | 87.8%

Bhopal, MP

Aug 2023 – Aug 2027

Rewa, MP

2022

TECHNICAL SKILLS

Languages: Python (Advanced), SQL (Advanced), R

ML / Statistical Modeling: Regression, Classification, Clustering, Random Forest, Hypothesis Testing, Probability & Quantitative Analysis

Deep Learning / NLP: CNN, BiLSTM, Attention Mechanisms, Transfer Learning, TensorFlow/Keras, NLP (Text Analysis, Feature Extraction)

Computer Vision: YOLO, OpenCV, Real-time Object Detection, Image Preprocessing

Data Engineering: Feature Engineering, ETL Pipelines, Data Cleaning, SMOTE, Big Data Concepts

Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, Excel (Advanced)

Databases: MySQL, MongoDB, SQL Query Optimization

Tools: Git, GitHub, Jupyter Notebook, Streamlit, VS Code, Google Colab

TECHNICAL PROJECTS

Stress Detection from ECG Signals | Python, CNN-BiLSTM, Attention, TensorFlow, WESAD

2026

- Developed an advanced statistical deep learning model (CNN-BiLSTM + Attention) for multi-class stress classification from physiological ECG signals, achieving **92%+ accuracy** and **94% recall**.
- Built a full data pipeline: R-peak detection, bandpass filtering, normalization, segmentation, and augmentation; applied **SMOTE** for class imbalance and **transfer learning** (MIT-BIH) for generalization.
- Executed rigorous model evaluation using Accuracy, F1-score, AUC-ROC, Precision, and Sensitivity — validating findings with statistical rigor for a healthcare-critical application.

Real-Time Traffic Monitoring & Analytics System | Python, YOLO, OpenCV, Data Pipelines

2025

- Designed and implemented an end-to-end data pipeline using YOLO and OpenCV to detect, classify, and count vehicles from live video feeds across multiple classes in real time.
- Built time-series aggregation and ETL logic to store vehicle count metrics, enabling traffic pattern analysis and supporting data-driven decision making from streaming sources.
- Optimized inference pipeline for near real-time throughput, demonstrating ability to handle large-scale, high-velocity data characteristic of Big Data environments.

YOLO Multi-Class Object Detection | Python, YOLO, OpenCV, Deep Learning

2024

- Built a real-time multi-class detection system achieving **25–40 FPS** on video streams; applied model optimization, image preprocessing, and feature engineering for maximum inference efficiency.

Exploratory Data Analytics & BI Dashboards | Python, Power BI, Tableau, SQL, Pandas

2023

- Conducted end-to-end EDA — data cleaning, feature engineering, statistical analysis, and hypothesis testing — delivering insights via interactive **Power BI** and **Tableau** dashboards for business stakeholders.
- Designed and optimized SQL queries for large-scale data extraction and transformation; reduced query execution time by **40%** through indexing and restructuring — demonstrating strong business acumen and quantitative reasoning.

CERTIFICATIONS

Neural Networks and Deep Learning | DeepLearning.AI / Stanford University (Coursera)

Feb 2026

Machine Learning Specialization | DeepLearning.AI / Stanford University (Coursera)

Oct 2025

Introduction to Structured Query Language (SQL) | University of Michigan (Coursera)

Sep 2025

ACHIEVEMENTS & LEADERSHIP

President, Tech Club

SAGE University Bhopal

Student Leadership

2024 – Present

- Led a 50+ member club organizing data science workshops, hackathons, and coding competitions; mentored 30+ peers on Python, ML pipelines, and data analytics lifecycle — fostering a collaborative, data-driven culture.
- Translated complex technical concepts into accessible peer sessions, demonstrating communication skills essential for cross-functional collaboration with business teams.

Student of the Year | Sunstone Bhopal — recognized for critical thinking, analytical skills & academic excellence